

HAIKANG DENG

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EDUCATION

University of California, Los Angeles

Ph.D. in Computer Science, Artificial Intelligence

Sep 2024 - Present

Advisor: Nanyun Peng

University of North Carolina, Chapel Hill

B.S. in Computer Science & Statistics, GPA: 3.96/4.0

Aug 2019 - May 2023

Advisors: Colin Raffel, Guanting Chen

PUBLICATIONS

- [1] Decoupling Task-Solving and Output Formatting in LLM Generation
Haikang Deng, Po-Nien Kung, Nanyun Peng
ACL, 2026. Oral Presentation
- [2] Learning Structured Reasoning via Tractable Trajectory Control
Po-Nien Kung, Zhen Yang, Jeffrey Luo, Cheng-Fu Yang, **Haikang Deng**, Zi-Yi Dou, Yinfei Yang, Nanyun Peng, Zhe Gan, Kai-Wei Chang
ICML, 2026. Spotlight
- [3] TaoBench: Do Automated Theorem Prover LLMs Generalize Beyond MathLib?
Alexander K. Taylor, Junyi Zhang, Ethan Ji, Vigyan Sahai, **Haikang Deng**, Yuanzhou Chen, Yifan Yuan, Di Wu, Jia-Chen Gu, Kai-Wei Chang, Nanyun Peng, Amit Sahai, Wei Wang
arXiv Preprint, 2026
- [4] DialectGen: Benchmarking and Improving Dialect Robustness in Multimodal Generation
Yu Zhou, Sohyun An, **Haikang Deng**, Da Yin, Clark Peng, Cho-Jui Hsieh, Kai-Wei Chang, Nanyun Peng
ResponsibleFM Workshop @ NeurIPS, 2025
- [5] Reward-Augmented Decoding: Efficient Controlled Text Generation With a Unidirectional Reward Model
Haikang Deng and Colin Raffel
EMNLP, 2023
- [6] Large Language Models Struggle to Learn Long-Tail Knowledge
Nikhil Kandpal, **Haikang Deng**, Adam Roberts, Eric Wallace, and Colin Raffel
ICML, 2023

EXPERIENCE

University of California, Los Angeles

Teaching Assistant

2025 - 2026

Los Angeles, CA

- CS 146: Introduction to Machine Learning (*Fall 2025, Winter 2026*)
- CS 263: Natural Language Processing (*Spring 2026*)

University of North Carolina, Chapel Hill

Research Assistant

Aug 2022 - Mar 2024

R-Three Lab

- Conducted research on human feedback alignment, controlled text generation, and retrieval augmentation
- Trained expert model routing and tagging in a contrastive manner with signals from a reward model

Amazon

Software Dev Engineer

Dec 2023 - Jul 2024

Bellevue, WA

- Designed automation for product pre-compute based on given marketplace, merchant, and store utilization score
- Composed a product lead time update tool that supports bulk updates (up to 1 million entries)
- Built a Local Landing Page to display local products available for pick up

RESEARCH PROJECTS

Analysis of RLVR's Effect on Model Reasoning Priors

Sep 2025 - Present

- Analyze distinct reasoning patterns in model-generated solutions to mathematical problems
- Cluster responses using reasoning chain logic and semantic similarities to correlate reasoning strategies with solution outcomes
- Quantify evolution of sequence probability, pattern frequency, and reasoning chain composition throughout RL training

Power Prover Models with Curriculum Learning

Jul 2025 - Present

- Benchmark prover model's performance on Terence Tao's Analysis textbook exercise problems
- Design a curriculum learning framework for prover models using a mixture of SFT and RLVR
- Generate synthetic training question through conjecturing and generating instantiations